Supplementary Material

Radiomics-Based Machine Learning and Deep Learning for Clinically Significant Prostate Cancer Detection on Biparametric MRI

S1 Sensitivity analysis: optimistic upper bound for the deep models under test-fold epoch selection

A natural concern when classical and deep models are compared is whether the deep architectures were disadvantaged by their training protocol. In the main analysis, the number of training epochs for each deep model was selected on an internal 80:20 split of the outer-training partition, after which the model was refit on the complete outer-training partition; the held-out outer fold was never used for any fitting or model-selection decision, exactly mirroring the leakage-safe treatment of the classical ensembles. To bound how much the deep models could gain from a more permissive—but optimistically biased—protocol, we repeated the deep-model experiments while selecting the training epoch by monitoring the held-out outer fold itself (up to 300 epochs). This procedure deliberately leaks information from the evaluation fold into model selection and therefore yields an *upper bound* on deep-model performance rather than a valid estimate of generalization; the classical models were left unchanged and serve as the leakage-safe reference. All metrics are pooled out-of-fold values with patient-level cluster-bootstrap 95% confidence intervals, reported at the fixed 0.5 probability threshold.

Even under this favorable protocol, the deep models did not overturn the main conclusions (Table S1). In the radiomics-only setting—the primary comparison—the leakage-safe Random Forest (AUROC 0.814) remained at least as discriminative as every test-fold-selected deep model (best deep AUROC 0.794), so the classical ensemble retained its advantage despite the deep models being granted a peek at the evaluation fold. Only in the combined conditions did the optimistically tuned deep models edge ahead of the Random Forest, and then only marginally and with broadly overlapping confidence intervals: by +0.009 AUROC for the concatenation Transformer (0.837 vs. 0.828) and by +0.015 AUROC for the dual-branch Transformer (0.843 vs. 0.828). These small, leakage-inflated margins are consistent with the main finding that, on this engineered radiomic feature space, deep tabular models do not confer a robust discrimination advantage over a calibrated tree ensemble, and they reinforce rather than weaken that conclusion: the deep models fail to clearly surpass the classical baseline even when the evaluation protocol is tilted in their favor.

Table S1: Optimistic upper bound for the deep models when the training epoch is selected on the held-out outer fold (a deliberately leakage-prone protocol), compared with the leakage-safe Random Forest reference carried over unchanged from the main analysis. Threshold-free metrics on pooled out-of-fold predictions with patient-level cluster-bootstrap 95% confidence intervals; lower Brier is better. Deep-model rows are optimistic upper bounds and are *not* valid estimates of generalization.

Condition	Model (protocol)	AUROC	AUPRC	Brier
Radiomics-only	Random Forest (leakage-safe, reference)	0.814 (0.790–0.837)	0.605 (0.561–0.655)	0.152
	Transformer (test-fold epoch)	0.794 (0.766–0.819)	0.585 (0.544–0.636)	0.161
	CapsNet (test-fold epoch)	0.761 (0.735–0.788)	0.559 (0.519–0.608)	0.202
	Transformer–CapsNet (test-fold epoch)	0.793 (0.767–0.818)	0.588 (0.545–0.634)	0.177
Radiomics+clinical (concat)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Transformer (test-fold epoch)	0.837 (0.815–0.860)	0.649 (0.606–0.697)	0.154
	CapsNet (test-fold epoch)	0.831 (0.809–0.853)	0.646 (0.605–0.691)	0.170
	Transformer–CapsNet (test-fold epoch)	0.828 (0.806–0.852)	0.652 (0.612–0.696)	0.166
Radiomics+clinical (dual)	Random Forest (leakage-safe, reference)	0.828 (0.805–0.851)	0.634 (0.591–0.683)	0.147
	Dual Transformer (test-fold epoch)	0.843 (0.821–0.864)	0.671 (0.628–0.716)	0.154
	Dual CapsNet (test-fold epoch)	0.840 (0.817–0.861)	0.661 (0.620–0.706)	0.151
	Dual Transformer–CapsNet (test-fold epoch)	0.829 (0.806–0.851)	0.621 (0.581–0.673)	0.171

S2 Radiomics-only per-model operating-point performance

Table S2 reports the full set of discrimination, calibration, and operating-point metrics for all seven radiomics-only models, complementing the forest plot of AUROC and AUPRC in the main text. Threshold-free metrics (AUROC, AUPRC, Brier) use the calibrated probabilities; sensitivity, specificity, balanced accuracy, F1, PPV, and NPV are reported at the per-fold outer-training-derived Youden operating point. Values are point estimates with patient-level cluster-bootstrap 95% confidence intervals for AUROC, AUPRC, sensitivity, and specificity (lower Brier is better).

Table S2: Radiomics-only model performance on pooled out-of-fold predictions.

Model	AUROC	AUPRC	Brier	Sensitivity	Specificity	Bal. acc.	F1	PPV	NPV
TabFM (foundation)	0.828 (0.805–0.850)	0.650 (0.600–0.699)	0.145	0.812 (0.774–0.848)	0.691 (0.664–0.718)	0.751	0.626	0.510	0.903
Random Forest (ML)	0.814 (0.790–0.837)	0.605 (0.556–0.658)	0.152	0.767 (0.724–0.806)	0.736 (0.710–0.761)	0.751	0.630	0.534	0.889
Gradient Boosting (ML)	0.802 (0.777–0.825)	0.592 (0.542–0.644)	0.156	0.774 (0.733–0.813)	0.675 (0.647–0.703)	0.725	0.597	0.485	0.883
Transformer (DL)	0.801 (0.775–0.824)	0.609 (0.559–0.662)	0.155	0.755 (0.714–0.795)	0.684 (0.657–0.711)	0.720	0.591	0.486	0.876
Transformer–CapsNet (DL)	0.803 (0.777–0.826)	0.600 (0.550–0.651)	0.154	0.807 (0.768–0.843)	0.672 (0.644–0.699)	0.739	0.612	0.493	0.898
LightGBM (ML)	0.792 (0.767–0.817)	0.598 (0.548–0.650)	0.159	0.791 (0.751–0.830)	0.672 (0.644–0.700)	0.731	0.603	0.488	0.890
CapsNet (DL)	0.794 (0.768–0.820)	0.589 (0.539–0.643)	0.155	0.753 (0.710–0.793)	0.685 (0.658–0.712)	0.719	0.590	0.486	0.875

S3 Model-agnostic SHAP attributions for the pretrained TabFM model

Because TabFM performs inference by in-context learning rather than through a differentiable forward pass over fixed weights, gradient-based attribution methods such as integrated gradients are not applicable. To characterize which features drove its predictions, model-agnostic permutation SHAP was instead computed for the pretrained TabFM model in the radiomics-only, clinical-only, and concatenation conditions. For each of the five outer folds, SHAP values were estimated on a held-out subsample of 25 validation studies against a 50-study background set, and the resulting per-study signed attributions were pooled across folds (Figure S1). The subsample size makes these attributions qualitative rather than a basis for fine-grained ranking, but they were computed with an attribution method independent of those used for the classical (SHAP) and deep (integrated gradient) models and therefore provide a cross-check on the shared feature signature. TabFM's leading features closely matched those of the other model families: diffusion-weighted texture and ADC first-order total energy dominated the radiomics-only condition, PSA density dominated the clinical-only condition, and in the concatenation condition PSA density and age rose above every radiomic descriptor—reproducing the re-ranking seen for the classical and deep combined models and confirming that the clinical variables supply largely independent risk information.

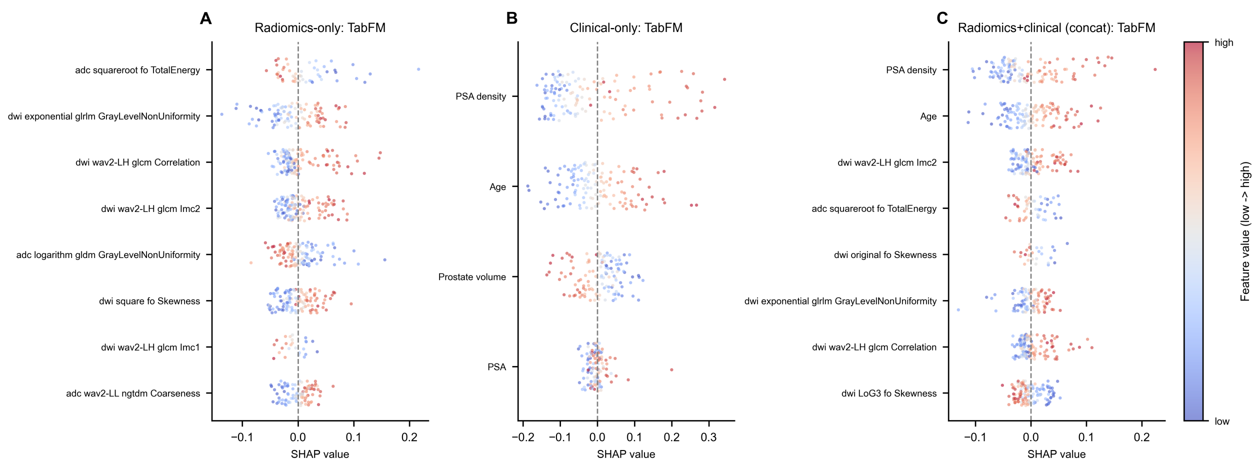
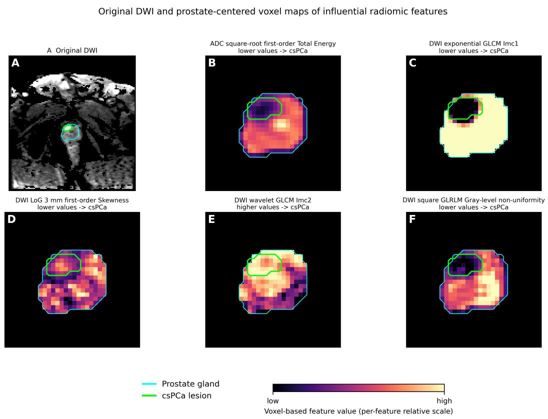


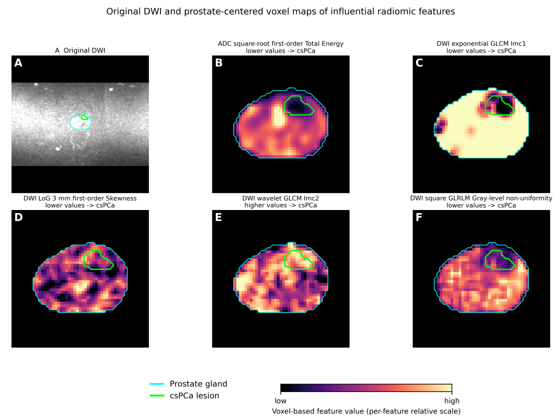
Figure S1: Model-agnostic SHAP attributions for the pretrained TabFM model. Beeswarm summaries of per-study signed permutation-SHAP values, pooled across the five outer folds' held-out subsamples, for the eight most important features in the (A) radiomics-only, (B) clinical-only, and (C) concatenation conditions. Each point is a study, positioned by its signed SHAP value and colored by the relative feature value (low to high); values to the right of zero increase the predicted csPCa probability. PSA density is the leading feature in the clinical-only and concatenation conditions, and diffusion-weighted texture with ADC first-order total energy leads the radiomics-only condition, matching the signature recovered by the classical and deep models with independent attribution methods.

S4 Voxel-based radiomic feature maps

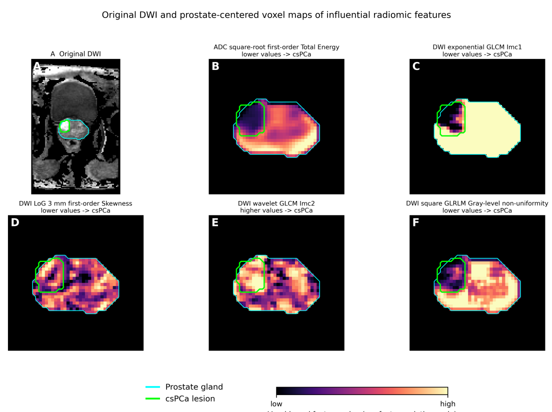
To provide case-level spatial explanations directly on the anatomy, a selected set of influential features was back-projected to the image as voxel-wise radiomic maps in csPCa-positive cases with human-expert lesion delineations (Figure S2). Their salient regions co-localized with the delineated csPCa lesion rather than being distributed uniformly through the gland, complementing the global importance summaries in the main text.



(a) Case 10322_1000328.



(b) Case 11377_1001400.



(c) Case 10383_1000389.

Figure S2: Original DWI and prostate-centered voxel-based radiomic maps in csPCa-positive cases with human-expert lesion delineations. In each subfigure, (A) original DWI slice with prostate gland (cyan) and csPCa lesion (green) delineations only; (B) ADC square-root first-order total energy, (C) DWI exponential GLCM Imc1, (D) DWI LoG 3 mm first-order skewness, (E) DWI wavelet GLCM Imc2, and (F) DWI square GLRLM gray-level non-uniformity. Feature panels show only the voxel map within a prostate-centered bounding box, with the gland and lesion contours retained. Colour encodes voxel-based feature value on a per-feature relative scale (low to high). Panel titles indicate the feature direction from the signed SHAP/integrated-gradient beeswarm analyses: lower values support the positive csPCa class for B–D and F, whereas higher values support the positive csPCa class for E.